

Lesson 2 - Practical Application Guide

□ How to Run the Project

```
cd D:\learnlaravel2025\lessons\lesson-02\practice-app
php artisan serve
```

Server will run on: `http://localhost:8000`

□ Available Routes

1. Home Page

- **URL:** `http://localhost:8000/`

2. Products Routes

- **GET** `/products` - Products list
- **GET** `/products/{id}` - Show specific product (numbers only constraint)
- **GET** `/products/{slug}` - Show product by slug

3. Admin Routes Group

- **GET** `/admin/dashboard` - Admin dashboard
- **GET** `/admin/users` - User management
- **GET** `/admin/products` - Product management
- **GET** `/admin/settings` - Settings

4. User Routes

- **GET** `/user/{name?}` - User page (optional name)
- **GET** `/profile` - User profile (named route)

5. Contact Form

- **GET** `/contact` - Display form

- **POST** /contact - Submit form
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📦 Implemented Exercises

Exercise 1: HTTP Methods Types

```
Route::get('/products', function () {
    return 'Products list';
});

Route::post('/products', function () {
    return 'Create new product';
});

Route::put('/products/{id}', function ($id) {
    return "Update product $id";
});

Route::delete('/products/{id}', function ($id) {
    return "Delete product $id";
});
```

Exercise 2: Route Parameters

```
// Required parameter with constraint
Route::get('/product/{id}', function ($id) {
    return "Show product number: $id";
})->where('id', '[0-9]+')->name('product.show');

// Optional parameter
Route::get('/user/{name?}', function ($name = 'Guest') {
    return "Hello $name";
})->name('user.greeting');

// Multiple parameters
Route::get('/category/{category}/product/{product}', function ($category, $product) {
    return "Category: $category - Product: $product";
});
```

Exercise 3: Named Routes

```
Route::get('/dashboard', function () {
    return 'Dashboard';
})->name('dashboard');

Route::get('/profile', function () {
    return 'User Profile';
})->name('profile');

// In Blade:
// <a href="{{ route('dashboard') }}">Dashboard</a>
// <a href="{{ route('product.show', ['id' => 5]) }}">Product 5</a>
```

Exercise 4: Route Groups

```
Route::prefix('admin')
    ->name('admin.')
    ->group(function () {
        Route::get('/dashboard', function () {
            return 'Admin Dashboard';
        })->name('dashboard');

        Route::get('/users', function () {
            return 'User Management';
        })->name('users');

        Route::get('/products', function () {
            return 'Product Management';
        })->name('products');

        Route::get('/settings', function () {
            return 'Settings';
        })->name('settings');
    });
```

Exercise 5: Form with GET & POST

```
// Display form
Route::get('/contact', function () {
    return view('contact');
})->name('contact.show');

// Process form
```

```
Route::post('/contact', function () {
    // Process data here
    return redirect()->route('contact.show')
        ->with('success', 'Your message has been sent successfully!');
})->name('contact.submit');
```

□ What We Learned

1. HTTP Methods Types

- GET for reading
- POST for creating
- PUT/PATCH for updating
- DELETE for deleting

2. Route Parameters

- Required parameters {id}
- Optional parameters {name?}
- Parameter constraints ->where()

3. Named Routes

- Easier route references
- ->name('route.name')
- route('route.name')

4. Route Groups

- prefix() - URL prefix
 - name() - Name prefix
 - middleware() - Shared middleware
 - group() - Group routes
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□ Useful Commands

```
# Display all routes
php artisan route:list

# Routes with specific prefix
php artisan route:list --path=admin

# Search by name
php artisan route:list --name=product

# Detailed view
php artisan route:list -v
```

□ Testing Routes

Visit each route to verify it works:

1. □ `http://localhost:8000/products`
2. □ `http://localhost:8000/product/5`
3. □ `http://localhost:8000/admin/dashboard`
4. □ `http://localhost:8000/user/John`
5. □ `http://localhost:8000/user` (without name)
6. □ `http://localhost:8000/contact`

□ Tips

1. **Use** `route:list` to display all routes
2. **Save changes** before refreshing browser
3. **Use named routes** always
4. **Organize routes** in logical groups
5. **Add constraints** to route parameters

□ Next Step

After completing this lesson, you're now ready for:

Lesson 3: Controllers and MVC Pattern

- Creating Controllers
- Organizing application logic
- Resource Controllers

Happy Learning! □